

Overthinking Detectors Transfer to New Models, Not New Tasks

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Abstract

Reasoning language models get more accurate when they are allowed to think longer. Spending extra compute this way at inference time is called test-time scaling (TTS). The benefit is not uniform. Sometimes the model keeps thinking after it has already reached a correct answer, and the longer chain ends on a wrong one. At best this wastes compute. At worst it flips a correct answer to wrong. This paper studies the harmful case. Existing monitors decide when to stop thinking from a single trigger signal, such as confidence or answer agreement. They are also built and tested on the same tasks. We ask what happens to such detection on models and tasks it has never seen. We measure overthinking per instance as a correct-to-wrong flip under longer thinking. We also build a reliability protocol that separates real flips from sampling noise. On the trusted labels, the detectable signature of overthinking belongs to the task, not the model. Task explains about two thirds of the signature variance. Model explains under one tenth. The reason is that the same cheap signal means different things on different tasks. Seven of eleven signals predict overthinking on one task family and predict its absence on another. The flipped signals include the trigger signals of existing monitors. A detector trained on some models therefore still works on model lineages it has never seen. That holds even for a lineage sharing neither pretraining nor post-training recipe. The same detector nearly stops working on tasks it has never seen. Deploying one trigger signal everywhere can make detection worse than random guessing. Our claim: an overthinking detector transfers to new models, not to new tasks. Read its signature per task, and reuse it across models. The limit is also measured. On GPQA our protocol certifies overthinking as real and dense, near 23% of instances. No cheap behavioral signal detects it there. Results are tested across about 2.1M graded generations: eight models, four pretraining lineages, and eight task families.

Introduction

Reasoning language models trained to “think before answering” improve markedly with more test-time compute. The gains are not uniform. On a sizable fraction of instances, *more* chain-of-thought turns a correct answer into a wrong one. We say the model **overthinks**: it second-guesses a right

answer, follows a distractor, or reasons its way to a wrong one (Chen et al. 2024; Gema et al. 2025; Zhou et al. 2026). If we could tell *in advance* which instances will overthink, we could stop early. We would then stop paying for compute that actively hurts. Overthinking is thus a concrete prediction problem at the heart of budget-aware inference.

The natural attack is to read cheap behavioral signals from a short prefix of the model’s reasoning. Those signals are token-level confidence, self-consistency across a few samples, and how often the model backtracks (“wait,” “hmm,” “let me reconsider”). A monitor learned on them flags likely overthinking. This is how the broader literature on uncertainty estimation and selective prediction builds monitors (Kadavath et al. 2022; Wang et al. 2023; Geifman and El-Yaniv 2017). Confidence and self-consistency are treated as properties *of the model*. They are calibrated once per model and then applied to every task. The implicit assumption is that *how to detect a failure* lives with the model, not with the task. We call this the **factorization assumption**. Confident-adaptive and early-exit systems (Schuster et al. 2022; Liu and Wang 2025) inherit it when they set a single confidence trigger per model.

Thesis. We show this assumption is **backwards for overthinking**. Detecting when a reasoning model overthinks depends on the *task*, not the model. The signature a monitor must read is set by the task and carries over to new models. So calibrate the monitor *per task* and reuse it across models. This is a claim about the detection *problem*, not about the cause. Model and task together set the *rate* of overthinking, which we do not contest. Our claim is about the *signature*. We establish it with no detector and no train/test split, by decomposing the variance of raw signal directions. It holds under three separately built labels, two of them denoised. Everything after that is consequence or limit. A trained detector transfers across models and collapses across tasks, forced by cues that flip sign between tasks. A control detector that does *not* collapse shows the limit of the effect. A pre-registered rerun with more samples per instance marks the limit of the approach: on the densest task (GPQA, ~23% of instances) the signature is uninformative. What remains we turn into a compute-accuracy curve, with the deciding seed kept apart from the scoring seed. Read whole, this is a measurement result. The label protocol is the instrument’s noise floor. The

task-determined signature is its calibration law. The GPQA detection failure is its measured operating limit. Fig. 1 shows the whole argument.

Contributions. (i) **The signature and its mechanism.** A detector-free variance decomposition shows the overthinking signature is set by the task (task share 0.66–0.79 vs. model 0.07–0.14 across three label constructions). The driver is that cheap signals change meaning across tasks. Under denoised labels, seven of eleven signals invert their overthink-direction across task families. This produces the cross-task collapse (0.61) while leaving cross-model transfer intact (0.84). (ii) **Label methodology and a measured boundary.** We give a reliability protocol for resampling-based overthinking labels: binomial nulls, cross-seed reproducibility, and a pre-registered $k=64$ resampling. It separates three regimes: labels that are noise, labels that are detectable, and labels that are dense but undetectable. On GPQA overthinking is dense and real yet undetected by all eleven signals. (iii) **A decoupled utility analysis.** We measure what the behavioral monitor gains over confidence baselines, and the detection gap that no deployable monitor closes.

Related Work

Overthinking and efficient reasoning. A growing line documents that longer chains can hurt and seeks to *reduce* overthinking via length penalties, adaptive CoT, distillation, or RL (Chen et al. 2024; Sui et al. 2025). Within this literature the word names three different quantities. Training-side work measures *waste*: the model keeps going after a verified answer has emerged, and sequence-level RL credit can reinforce this (Wei et al. 2026). Waste rarely changes correctness. Compression work measures a *regime*: a negative group-wise correlation between length and correctness (Wang et al. 2026). We measure *harm*: a per-instance accuracy flip under added budget, with a definite sign. Whether the three coincide is unmeasured, and results do not transfer between them. Closest to us, concurrent work (Zhou et al. 2026) tracks marginal utility and correct→wrong flips across budgets, but with population-level marginal utility, two models, single greedy decodes, no treatment of label noise, and indicator correlations only in-distribution on AIME. We differ in *what is measured*: per-instance labels under an explicit reliability protocol, plus the structure of detectability, which is task-determined, model-portable, and usable by any mitigation method.

Inverse scaling and test-time compute. The Inverse Scaling Prize first catalogued tasks where larger models do *worse* (McKenzie et al. 2023). Test-time-scaling studies report average gains and a slice where accuracy drops (Snell et al. 2024; Muennighoff et al. 2025). Gema et al. (2025) *construct* tasks on which more thinking reliably lowers accuracy, and catalogue failure modes by model family. Our suite overlaps theirs, but we ask a complementary question: can a cheap monitor *predict which* instances suffer, and what does that prediction depend on? We find the predictor is bound to the task, a structural claim their construction-oriented study does not make. This is consistent with their catalogue. Model family moves the *rate*, and Phi-4 is genuinely elevated in our data,

but a two-way decomposition of the rate is itself dominated by the task. Rate and signature are different objects.

Task topology. Tong, Cakar, and Pehlevan (2026) relate length-generalization and the benefit of reasoning traces to a task’s *topology* (depth vs. breadth). Reasoning helps on broad-shallow tasks and hurts on narrow-deep ones, which supports our claim that task structure, not the model, governs when extra reasoning lowers accuracy. We extend it from *when reasoning helps* to *what a monitor must read to detect when it hurts*.

Uncertainty, calibration, self-consistency. Confidence (Kadavath et al. 2022), self-consistency (Wang et al. 2023), and selective prediction (Geifman and El-Yaniv 2017) treat their signals as model-level and task-general. We show that for overthinking they change sign from one task to another, while our control shows they stay task-general for plain error. Closest to us, Orgad et al. (2025) show that *internal-state* error probes are dataset-specific. Ours is the behavioral analogue and sharpens theirs twice over: the error direction *inverts* rather than merely failing to transfer, and even our task-robust wrong-answer control drops from 0.722 pooled to 0.59 macro-averaged across tasks.

Adaptive computation / early stopping. CALM (Schuster et al. 2022), answer-convergence stopping (Liu and Wang 2025), and certainty-based serving (Fu et al. 2024) all need a per-instance trigger. The standard recipe calibrates one confidence threshold per model. We show the trigger should instead be set per task, and that it carries over across models. We also quantify what that buys on a mixed-task workload.

Method

Fig. 1 sketches the argument end to end. The rest of this section gives the measurement machinery it relies on.

Scale and leakage-free measurement. Every number below is computed over $\sim 2.1\text{M}$ graded generations: 8 models $\times$ 8 task families $\times$ 7 thinking budgets $\times$ ($k=8$ samples + greedy) $\times$ 3 seeds. Two choices keep the labels from being inflated. (i) **Grade on the full output.** RL-style models often write the final `\boxed{\}` answer *inside* `<think>` and only paraphrase it afterward. Grading the post-think text alone misses that answer and records a false long-budget “error.” We grade the *full* output (thinking + answer). We take the last `\boxed{\}` or “Final answer” anywhere in it, computed in process on untruncated text. (ii) **Per-task answer caps**, set by a saturation sweep: 64 tokens for MC, 256 for math, 512 for word problems and logic, 1024 for code. Truncation then never clips a genuine answer.

Overthinking, measured without constructed labels. For each problem with a programmatic gold answer we elicit reasoning at thinking budgets b by truncating the chain of thought at a token cap and forcing an answer. From $k=8$ samples at temperature 0.6 plus one greedy decode we estimate $P_{\text{correct}}(b)$ and define the marginal utility of extra thinking

$$\text{MU} = P_{\text{correct}}(b_{\text{long}}) - P_{\text{correct}}(b_{\text{short}}),$$

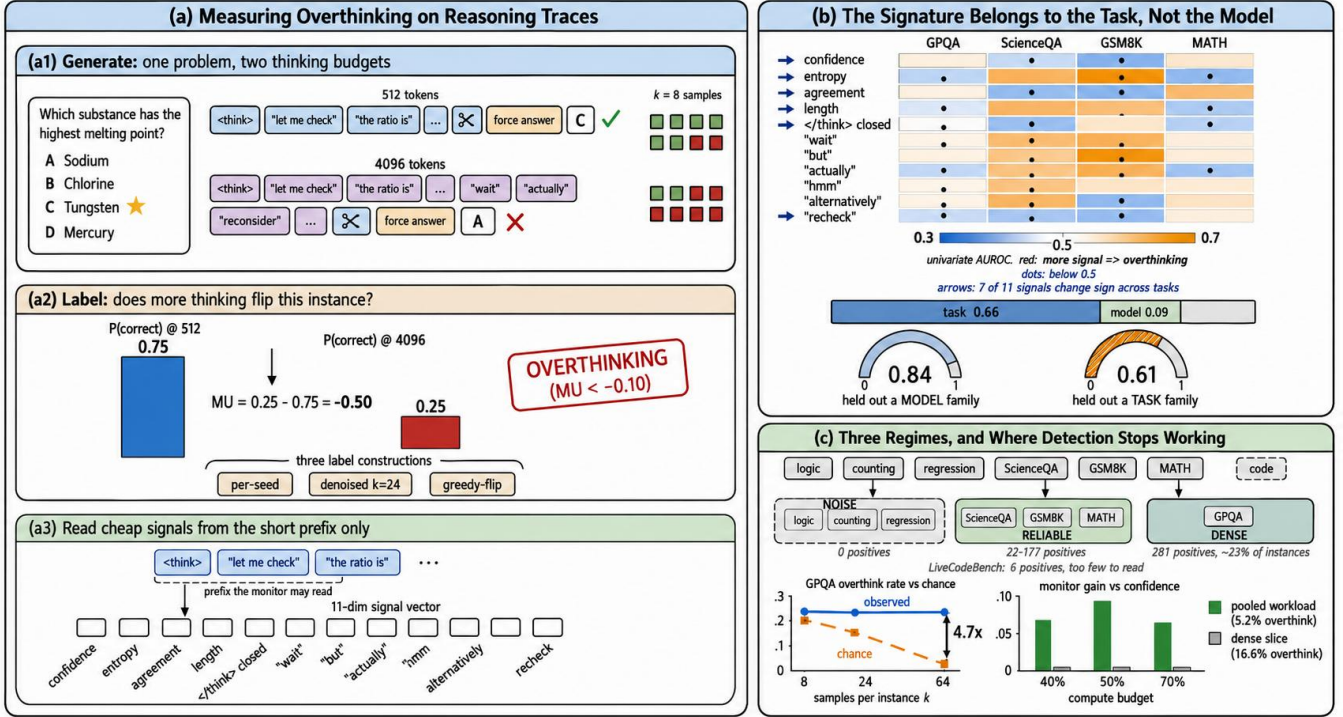


Figure 1: **(a)** An instance is overthinking when the longer budget lowers its graded accuracy by more than $\tau=0.10$. A monitor reads only the short prefix. **(b)** Each cell is one signal’s univariate AUROC on one task family (denoised label); red: more signal goes with more overthinking. Seven of eleven signals cross 0.5, and the GPQA column is uniformly pale. **(c)** Three families are retracted as sampling noise; GPQA is dense yet its monitor gain is zero.

with $b_{\text{short}}=512$ and $b_{\text{long}}=4096$. We call an instance **overthinking** iff $MU < -\tau$ ($\tau=0.10$). The label is per instance and specific to the model. It is measured, not constructed. It answers one question: does more thinking flip this answer away from the gold? We define difficulty as P_{correct} (2048). We show below that it is *not* a cheap feature one can deploy.

Three label constructions and a policy. From $k=8$ samples per seed at each budget we form three labels. The *per-seed* label thresholds MU within each seed ($\tau=0.10$). Its coverage is the largest of the three, but it is quantized at $1/8$, so a single net sample can flip it. The *denoised* label pools all three seeds’ 24 samples per budget and thresholds the pooled MU. It covers only instances with all three seeds present. Quantized at $1/24$, it removes most single-sample flips. The *greedy-flip* label marks an instance when the greedy decode is correct at 512 and wrong at 4096 in at least two of three seeds. It shares no sampling machinery with MU, so it independently checks how the label is built. All structural results are reported under the *denoised* label, with the other two as robustness. The utility simulation keeps *per-seed* labels, because it calibrates on two seeds and scores on the held-out third. Positive counts: denoised 554 (GPQA 281, ScienceQA 177, math 68, GSM8K 22, code 6, others 0), greedy-flip 315, per-seed 2747.

Models, tasks, signals. *Models* (8): R1-Distill-Qwen {1.5,7,14,32}B and QwQ-32B (Qwen2.5 base; distill and

RL) (DeepSeek-AI et al. 2025; Qwen Team 2025), R1-Distill-Llama-8B (Llama-3.1), Phi-4-reasoning-14B (Abdin et al. 2025), and OLMo-3-7B-Think (OLMo/Dolma, Ai2 recipe; held out for breadth) (OLMo Team 2025). The five R1-Distill models are DeepSeek-R1 distills. *Tasks* (8): MATH-500 (Lightman et al. 2024) plus AIME, GPQA-Diamond (Rein et al. 2024), GSM8K (Cobbe et al. 2021), ScienceQA (Lu et al. 2022), LiveCodeBench (Jain et al. 2025) (graded by executing against public tests in a sandbox), and three families we generate: knights-and-knaves logic, counting-with-distractors, and linear regression with spurious features. *Signals* (read at the short budget): mean answer log-probability (confidence), self-consistency entropy and top-answer agreement, mean thinking tokens, fraction of samples closing $\langle \text{think} \rangle$ naturally, and the rates of six backtracking markers. All runs use three seeds, 31, 1234 and 2026; we report mean $\pm$ std. Generation ran one job per NVIDIA H200 GPU (8 CPU cores, 100 GB host memory, Rocky Linux 9.7) under vLLM 0.11, PyTorch 2.8 and Python 3.12; analysis used scikit-learn 1.9, NumPy 2.2, pandas 3.0 and SciPy 1.18.

Protocol. The detector is a logistic regression on the standardized signal vector with balanced class weights. In distribution we report pooled out-of-fold cross-validated AUROC. For transfer we report *leave-one-out* AUROC on two axes: leave one task family out, and leave one model base family out. The features are the same task-agnostic ones on each axis. The two legs are evaluated on *different* populations, so

we do not read the raw gap between them as a like-for-like comparison. A matched-population analysis in the appendix isolates the true difference between the axes. The signature analysis fits nothing. Per (model, task) cell it computes the vector of *univariate* AUROCs of the signals, then correlates these vectors across the model axis and across the task axis. It is a purely descriptive statistic.

Which Overthinking Labels Can Be Trusted?

We build the argument in two acts. First we ask which overthinking labels can be trusted at all. We check each family’s overthink rate against sampling noise, and rerun the unclear families at $k=64$ as pre-registered. The eight families sort into three regimes: noise, reliable, and dense (this section). Only on the trustworthy regimes do we then establish the structural claim and translate it into utility. P_{correct} is estimated from $k=8$ samples per seed. So MU is quantized at $1/8$, and at $\tau=0.10$ the per-seed label flips on *one net sample*. We measure that noise and remove it. The picture after removal is more interesting than the picture before it.

A per-family noise diagnostic. For each family we compare the observed per-seed overthink rate to a binomial equal- p null. Table 1 reports three further quantities: cross-seed Jaccard overlap, per-instance one-sided Fisher-exact tests on the 3-seed pooled 24-vs-24 counts (BH-FDR 0.05), and the positive counts under the denoised and greedy-flip labels. At $k=8$ the null is very wide, since $\tau=0.10$ is one net sample. Every family’s observed rate is then at or below its null. The diagnostic can identify noise but cannot certify signal. Three families are unambiguous even so. Logic, counting and regression have zero positives under both denoised labels and Jaccard ≤ 0.007 . Their per-seed positives are sampling noise, and we retract them below. The cues that that these families supported are exactly the four that stop flipping under denoising (next section). The Fisher tests find 95 individually significant instances. They concentrate in the science families (68 GPQA, 15 ScienceQA, 7 GSM8K, 5 MATH, zero elsewhere).

A pre-registered $k=64$ resampling. The two dense families are *at* their $k=8$ null. That allows two readings: the positives are inflated by noise, or the small-sample null is too wide to resolve real density. We pre-registered both readings and reran GPQA and ScienceQA at $k=64$ (three models, both budgets, three seeds). The second reading is correct. GPQA’s overthink rate is stable in k ($0.259 \rightarrow 0.239 \rightarrow 0.232$ for $k=8/24/64$) while its null falls ($0.236 \rightarrow 0.050$). At $k=64$ the observed rate is $4.7\times$ its null, uniformly across models ($3.9\text{--}5.3\times$). GPQA overthinking is real and dense, $\sim 23\%$ of instances. ScienceQA shows the same pattern at lower density (rate 0.081, $2.9\times$ null). But its density varies across models (obs/null 1.4 on the 7B up to 4.7 on OLMo), so we scope the dense-regime claim to GPQA.

The regime map. Denoising separates the eight families into three regimes. The *noise* regime is logic, counting and regression: their per-seed positives are absent under both denoised constructions, and we retract them from every claim that follows. The *reliable mid-band* is ScienceQA, GSM8K

Table 1: Label-SNR diagnostic (per-seed $k=8$ rate and equal- p binomial null; cross-seed Jaccard; Fisher-significant instances at BH-FDR 0.05; denoised $k=24$ positives), plus the pre-registered $k=64$ resampling for the two science families. Per-model $k=64$ GPQA obs/null: 7B $4.7\times$, 14B $5.2\times$, OLMo-7B $3.9\times$.

family	rate	null	Jac.	#Fis.	n_{24}	k_{64} o/n
GPQA	0.263	0.256	0.415	68	281	4.65
scienceqa	0.133	0.133	0.299	15	177	2.85
math	0.050	0.246	0.173	5	68	–
gsm8k	0.018	0.127	0.261	7	22	–
livecodebench	0.036	0.220	0.103	0	6	–
counting	0.004	0.105	0.000	0	0	–
logic	0.010	0.334	0.006	0	0	–
regression	0.000	0.330	0.000	0	0	–

and MATH: denoised positives in the tens to hundreds, stable across seeds. The *dense* regime is GPQA: certified real at $4.7\times$ its null, $\sim 23\%$ of instances. All structural claims are stated on the mid-band and the dense regime. Whether the certified overthinking can actually be *detected* differs sharply between them, as the next section shows.

The Overthinking Signature Is Task-Determined

A Detector-Free Variance Decomposition

We state the main result in a form that cannot beg the question: *no detector, no train/test split*. For each (model, task) cell we summarize how overthinking *looks* by an 11-dimensional **signature**. Each entry is the univariate AUROC of one signal. Above 0.5 means more signal goes with more overthinking, and below 0.5 the opposite. Do these vectors cluster by model or by task? They cluster by task. We use the denoised label on the cells with ≥ 10 positives (15 cells, 3 families). The mean signature correlation *within a task, across models* is **0.37**. *Within a model, across tasks* it is **0.06**. A two-way type-II sum-of-squares decomposition gives **0.66** of the signature variance to task and **0.09** to model. Under the per-seed label the split is 0.72/0.07, and under the greedy-flip label 0.79/0.14. The direction does not depend on how the label is built. A 1000-cell bootstrap gives task 95% CI [0.17, 0.77] vs. model [0.06, 0.36] under the denoised label. $P(\text{task} \leq \text{model})$ is 0.072 denoised, 0.000 per-seed and 0.025 greedy-flip, so ≤ 0.07 under every label (Fig. 2). The task determines most of what overthinking looks like. The model determines almost nothing. Every downstream transfer result follows from this structure, not from a fitted detector.

A Detector Transfers Across Models, Not Across Tasks, and Why

Under leave-one-out, holding out an entire **model base family** costs little. Holding out a **task family** is severe. Under the denoised label, cross-model out-of-distribution AUROC is **0.84** and cross-task is **0.61**. The per-seed label gives 0.77 vs.

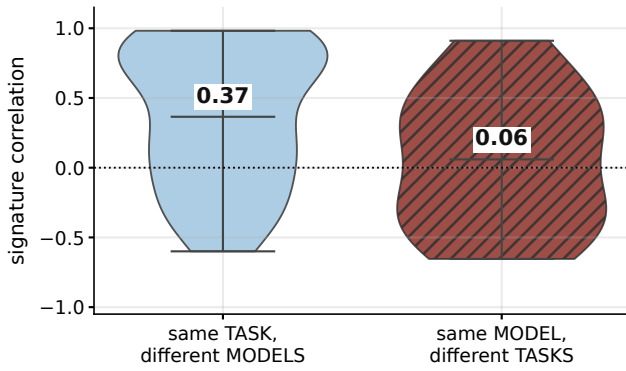


Figure 2: **The signature clusters by task, not model** (denoised label, no detector, no train/test, 15 cells). Pairwise correlations of the 11-dim signature: across models within a task (mean 0.37) vs. across tasks within a model (mean 0.06, centred on zero).

Table 2: Transfer asymmetry under all three labels (leave-one-out OOD AUROC, min-8-positive floor). Cross-model exceeds cross-task by 0.18–0.24 every time.

label	pos.	cross-task	cross-model	asym.
per-seed	2747	0.520	0.765	0.245
denoised	554	0.606	0.845	0.238
greedy-flip	315	0.592	0.775	0.183

0.52 and the greedy-flip label 0.78 vs. 0.59. The gap is 0.18–0.24 under every construction (Table 2). Per held-out task under the denoised label, no family *inverts* (Fig. 3). GPQA (0.51) is at chance and math (0.58) is weak, while GSM8K (0.66) and ScienceQA (0.68) keep signal. Counting, logic and code no longer appear, since their denoised positives fall below the floor. So the earlier per-seed inversions (code 0.35, counting 0.38) are not denoised phenomena. A “narrow detector” null predicts that *both* axes fail. Only the task axis does.

Mechanism: cheap signals change meaning across tasks.

A detector *cannot* be task-general, because the cues flip meaning across tasks. We call this **task-polysemy**. Under the denoised label, **seven of the eleven signals cross** 0.5 (Fig. 1b). Each of the seven predicts overthinking in one task family and its *absence* in another. All seven reproduce under the independent greedy-flip label. Self-consistency entropy predicts overthinking on GSM8K (0.71) but inverts on competition math (0.38). Answer agreement runs the other way (MATH 0.63, GSM8K 0.29). The “actually” backtracking marker flips between the same pair (GSM8K 0.72, MATH 0.43). Three of the seven cross 0.5 only marginally on one side, within 0.03. The unambiguous flips on both sides are self-consistency entropy, agreement, and the “actually” marker. The remaining four flipped only under the noisy per-seed label, where all eleven flip. Each traces to a single family whose positives disappear under denoising. The denoised flip set is smaller and cleaner, not weaker. No

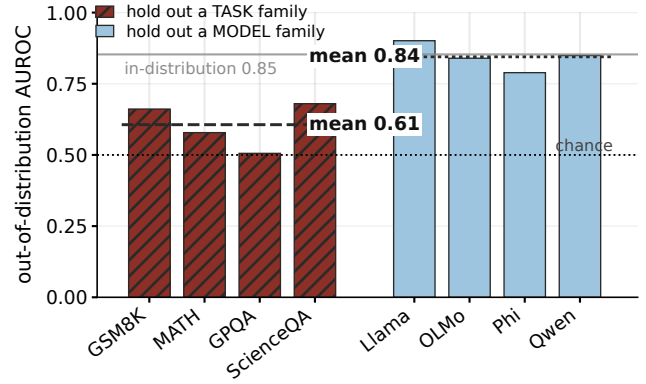


Figure 3: **Transfer is asymmetric, family by family** (leave-one-out OOD AUROC, denoised). Red: held-out task families lose signal, GPQA reaches chance. Blue: held-out model families stay near the in-distribution 0.853.

single linear rule can fit opposite signs, so the cross-task collapse follows from the data. Within a task the signature does port to unseen model families: a per-task detector trained on source families and tested on a held-out one scores 0.69 on ScienceQA and 0.65 on GSM8K. The appendix explains why each remaining cue flips.

Breadth, and a teacher-lineage control. A genuinely new lineage confirms the model half. Training on the other three families and testing on *OLMo* (Dolma pretraining, Ai2 recipe) yields out-of-distribution AUROC **0.805 ± 0.015**. *OLMo*’s own predictor reaches 0.811, and per-model predictors span 0.76–0.84. Five of the eight models are DeepSeek-R1 distills. So “transfers across models” could really mean “transfers across R1-recipe variants.” It does not. A detector trained *only* on R1-distill models ports to the independent-recipe targets at AUROC 0.809 (*OLMo*) and 0.808 (*QwQ*), with Phi the outlier at 0.639. The reverse direction gives 0.75–0.81. The sharpest test is the **QwQ↔OLMo** pair, which shares *neither* base pretraining *nor* post-training recipe. It still transfers at 0.805/0.806. Sharing a recipe adds little. The model half is a real cross-lineage property, with Phi the one weak cell.

Pooled AUROC reports task identity; read detection within task.

Positives concentrate in two families. So a detector that reads *only* task identity scores **0.886** pooled under the denoised label, above the pooled logistic at 0.853. Pooled numbers mostly reflect *which task*. The honest per-instance quantities are within task, and there the bar is not 0.5. Models differ in overthink rate inside a family, so the reference for a behavioral signal is a model-identity prior. The signals score well above that prior on ScienceQA (0.835 vs. 0.656) and GSM8K (0.801 vs. 0.615), and slightly above on MATH (0.621 vs. 0.585). On GPQA they are at chance (0.534) and fall *below* the model prior (0.570). The per-seed label gives the same result (0.540 vs. 0.557). GPQA is not a sparse-label artifact. The regime map certifies its overthinking dense and real, yet here it is at chance. This is the boundary we return to at the close. Table 3 reports all columns. A matched-

Table 3: **Within-task** detectability against a *model-identity prior* (denoised label; n_{24} : denoised positives). The bar is the per-model overthink-rate prior, not 0.5. On GPQA the signals fall *below* it: dense yet undetectable. code[†] has only 6 positives; logic, counting and regression have *zero*.

task	n_{24}	signal	prior	+model
ScienceQA	177	0.835	0.656	0.852
GSM8K	22	0.801	0.615	0.813
math	68	0.621	0.585	0.588
GPQA	281	0.534	0.570	0.531
code [†]	6	0.716	0.835	0.834

population analysis in the appendix shows the gap between the axes is real but smaller like-for-like (≈ 0.07), which restates the thesis.

The Collapse Is Specific to Overthinking

Is task-locking just what our pipeline does to everything? No. We ran the identical pipeline on a different target: detecting a *wrong answer* from confidence, self-consistency and cross-budget disagreement. It transfers across models (0.81) and across tasks (0.722). A wrong answer has a low-confidence profile that is nearly universal. Individual error cues do still flip on some tasks (4 of 5 by the same criterion), but the dominant confidence signal does not. The multivariate error detector therefore ports across tasks anyway. The overthinking detector has no dominant stable cue (7 of 11 cues flip denoised, 11 of 11 per-seed) and collapses to 0.52 per-seed. The collapse is specific to overthinking. It is not a property of all reasoning-failure detection (appendix).

Is the Detector Just Difficulty? A Cheap-Difficulty Defense

Split the features apart and difficulty *is* the component that transfers. A difficulty-only detector transfers across *both* axes (cross-model 0.74, cross-task 0.69). A signals-only detector transfers across models (0.77) but not tasks (0.52). The catch is that this difficulty is $P_{\text{correct}}(2048)$, which costs the very budget early stopping aims to save. The honest question is whether a *cheap* difficulty, read from the ≤ 512 prefix, also transfers. It does not. $P_{\text{correct}}(512)$ detects overthinking at 0.633 in distribution and transfers across tasks at only 0.620, against 0.689 for the expensive P_{2048} . Difficulty *predicted* from the short signals collapses to **0.522** cross-task. The signals are also not redundant with it. Residualizing the 11 signals on cheap difficulty still leaves AUROC 0.785, against 0.633 for cheap difficulty alone. The union reaches 0.813. The tightest claim has two halves. The cause that transfers across tasks is difficulty, and it is not cheaply observable. The cheaply observable symptom is the behavioral signature, and it is locked to the task. Either way the online detector must be calibrated per task.

Does It Buy Anything? A Utility Simulation

AUROC is not the deliverable. Saved compute is. We simulate an early-stopping controller over the pooled multi-task work-

load (six full-budget models, three seeds). For each instance a monitor chooses STOP-AT-SHORT (take the 512 answer) or PAY-FOR-LONG (take the 4096 answer). Expected accuracy is the mean P_{correct} of the chosen budget. Expected cost is the mean thinking tokens. One knob, the fraction stopped short, traces each monitor’s compute–accuracy frontier. **Decision and evaluation are decoupled.** A naive simulation lets a policy decide and be *scored* from the same $k=8$ estimate. That inflates exactly the policies whose decision feature is a graded label. We remove it by calibrating every monitor on two seeds and scoring on the held-out third, averaged over the three splits. The oracle is calibrated the same way. We compare the *always-long* and *always-short* anchors, a per-model **confidence monitor**, a **behavioral monitor** (logistic on the 11 signals), a **deployable cheap-difficulty router** that uses a label-free P_{512} predicted out of fold from the signals, and a **held-out MU-oracle**.

1. **Behavioral signals carry real early-stopping value over a confidence threshold.** At every operating point and calibration level the behavioral monitor scores above the confidence monitor: at 50% compute, 0.718–0.759 vs. 0.664–0.689 (Fig. 4). The 11 cheap signals are worth more than answer log-probability alone.
2. **The gain one can actually realize is modest, because overthinking is rare (5.2%).** The held-out MU-oracle peaks at 0.877 against always-long 0.870, a ceiling of +0.7 points, and at equal accuracy cuts compute to 77%. (An earlier same-seed simulation reported +1.4 points and 66%; roughly half was optimism from letting one seed both decide and score.) The gain is compute saved, not accuracy gained.
3. Per-task calibration helps *detection*, not the pooled *routing* frontier. With features fixed, a per-task *quota* is the worst router (0.718 at 50% compute) and one global policy the best (0.759), because pooled routing is a cross-task *allocation* problem. Reading the signature per task while allocating with one *global* threshold recovers the loss (0.775). Detection is a different object: there the per-task reading gives AUROC 0.78 against 0.52 cross-task.
4. A deployable cheap-difficulty router is *not* a strong baseline. Once P_{512} is predicted out of fold and scored on a held-out seed it saves essentially nothing at equal accuracy, and scores below the behavioral monitor at every budget $\leq 70\%$ (0.714 vs. 0.759 at 50%; 0.806 vs. 0.826 at 70%).
5. **Where overthinking is densest, no deployable monitor realizes the headroom.** On the overthink-dense slice (rate 16.6% vs. 5.2% pooled) the oracle’s headroom is largest, at +2.3 accuracy points and 39% of always-long compute. But the behavioral monitor’s edge over confidence collapses to ≤ 0.5 points at every budget (0.03/0.49/0.39 at 40/50/70%), and its peak accuracy merely matches always-long (0.7175 vs. 0.7170). The slice is dominated by GPQA, where the signature is uninformative; the complementary low-overthink slice behaves like the pooled workload (behavioral 0.757 vs. confidence 0.672 at 50%). The largest gain is exactly where cheap behavioral detection fails.

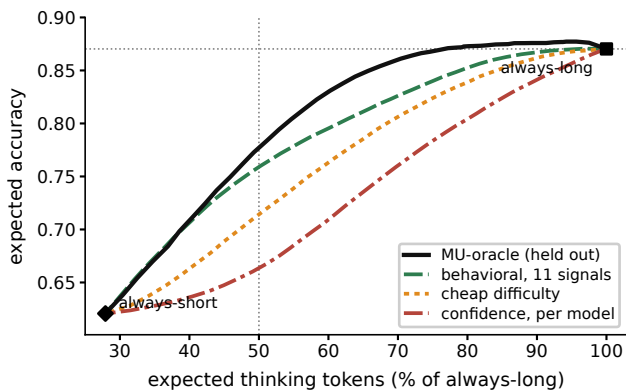


Figure 4: **Early-stopping utility** (pooled workload; decision seed decoupled from scoring seed). The behavioral monitor (green) scores above the confidence baseline (red) everywhere, and above the deployable cheap-difficulty router (orange) at every budget $\leq 70\%$. All sit below the held-out MU-oracle (black, $+0.7$ -pt ceiling).

Three recommendations follow. Read the detection *signature* per task. For routing, the effective lever is allocating compute across tasks by expected overthink rate. On mixed workloads a behavioral monitor scores above a confidence threshold at every budget. On the dense slice no cheap monitor currently realizes the oracle’s headroom.

Robustness

All checks are re-analysis, not new generation; the supplement has the detail. The model-versus-task gap holds three ways. It is robust to τ across $\tau \in \{.05, .10, .15, .20\}$ (model $0.77 \rightarrow 0.83$, task $0.52 \rightarrow 0.60$). It is threshold-free: the best [Spearman(signal, MU)] is uniform across models at the $0.59\text{--}0.70$ but swings across tasks at $0.11\text{--}0.73$. It is not sparse-task noise, since the denoised label still gives cross-task 0.61 against cross-model 0.84 . A bootstrap gives a pooled model-mean 95% CI of $[0.756, 0.777]$ against a task-mean of $[0.494, 0.543]$, non-overlapping, with the task interval including chance.

Discussion and Conclusion

The model-portability we claim is real but imperfect: held-out-family AUROC averages 0.84 denoised and 0.77 per-seed, with a worst cell of 0.65 (Phi). For routing that suffices, because calibration is what matters. Getting the per-task *signal* wrong is not merely suboptimal. It is *below chance*. Routing each task to its best single feature lifts pooled detection AUROC to **0.73** , one global feature gives **0.57** , and the worst per-task choice gives **0.48** ($0.72/0.58/0.45$ denoised, so the below-chance case survives denoising). A practitioner who picks one signal model-wide can do actively harmful detection. The more serious caveat is not rarity: where overthinking *is* common, our stress-slice simulation shows the behavioral monitor’s edge disappears, because the dense regime is exactly where the signature is uninformative. The natural next step is from behavioral signals to internal ones.

Can error probes that read hidden states (Orgad et al. 2025) detect the GPQA overthinking that behavior cannot? That is the sharpest question we leave open.

Overthinking is a **task-shaped** failure. Who overthinks how much is set by model and task together. But what overthinking *looks like to a monitor* is the task’s own signature, and it barely moves with the model. It transfers zero-shot even to an unseen pretraining lineage, and it holds under every way we build the label. The mechanism is that the same cheap cue means overthinking on one family and healthy reasoning on another. So no task-agnostic monitor can work, while one per-task monitor can be reused across models. Read the detection signature per task and reuse it across model families. This inverts the model-centric default of confidence-based monitoring, and getting it wrong scores below chance. The approach has a measured boundary. On the densest task, certified real by resampling at $k=64$, no cheap behavioral signal detects it, and the largest early-stopping gains are not realized. Detection is task-specific. Its failure region is where the problem is densest. Both are measured, not modeled. The open question is whether training choices, meaning data, algorithm and on- versus off-policy signal, manufacture the overthinking measured here, relocate it, or merely amplify it.

Limitations

Three families (Llama, Phi, OLMo) contribute a *single* model each, so their leave-one-base-family-out scores are single-model generalization tests; only Qwen contributes several, and Phi is the weakest held-out cell (0.65). All eight models are $1.5\text{--}32\text{B}$ open R1-style thinkers and five are DeepSeek-R1 distills. A teacher-disjoint control (QwQ $\leftrightarrow$ OLMo 0.805) shows the transfer is not a shared-recipe artifact, but frontier and closed models remain untested. GPQA’s within-task signature neither detects nor ports (0.48 , per-seed). The utility ceiling is modest because overthinking is rare on strong models and easy tasks, so our compute savings are conservative. Overthinking is operationalized at one $(\tau, b_{\text{short}}, b_{\text{long}})$ choice, with τ robustness shown but not exhaustively. The utility simulation uses one threshold per calibration group and charges only the chosen rollout’s budget, not the shared cost of the short-prefix signals, so absolute savings for multi-sample monitors are optimistic. The denoised label restricts structural claims to the four families with ≥ 20 positives; GSM8K’s cell is small (22) and MATH’s within-task detectability drops under denoising ($0.705 \rightarrow 0.621$), so mid-band magnitudes carry wider error than the pooled numbers suggest. The $k=64$ certification covers three models and the two science families at about $8\times$ the sampled-arm compute of the $k=8$ grid; ScienceQA’s density there varies across models, so the dense-but-undetectable claim is scoped to GPQA. The matched-population analysis is per-seed-label only (22 matched cells, below our 40 -cell floor).

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